Easing Elevator Congestion in a Vertical Classroom Building: A Simulation Study of Mundelein Center

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Abstract

Mundelein Center, on Loyola University Chicago's Lakeshore Campus, holds 54 general-purpose classrooms on twelve floors but is served by only two elevator banks: four cars that reach floor 9 and two cars that reach floor 14. Each class changeover produces long lines. We build a discrete-time simulation of one 15-minute passing period and use it to compare two low-cost behavioral measures: holding backpacks in front, and walking to and from floors 2 and 3. We also test whether running floors 10–14 on a time offset spreads the load. In the baseline configuration, mean waiting time is 5.2 minutes and 23% of students arrive after class has started. Backpack compliance cuts mean wait by 31% and nearly halves lateness; walking removes only 6% of the wait; together they cut mean wait by 35% and lateness to 11%. Offsetting the upper floors by 5–10 minutes shortens the time to clear the queue by about 20% but, combined with the other measures, *raises* the share of late students. The qualitative ordering of the measures is stable across occupancy, door time, and adjacent-floor traffic. The absolute numbers are not measurements: every timing parameter is an assumption, and we close with the timing study needed to calibrate the model.

1 Introduction

Mundelein Center is a tall, narrow classroom building. Its classrooms are spread from floor 1 to floor 14, and all of them are reached through two elevator banks of four and two cars. Only the two-car bank travels above floor 9. Every 75 minutes on Tuesdays and Thursdays and every 50 minutes on Mondays, Wednesdays, and Fridays, students leaving one class and arriving for the next converge on the elevators. Cars leave the upper floors carrying four to six people and still stop at every floor below, even though they are physically full, so students on the lower floors wait for cars that cannot take them.

We focus on classes that end before 4 PM and on two behavioral measures that cost nothing to implement: asking students to take their backpacks off and hold them in front (our estimate: two more riders per car), and asking students on floors 2 and 3 to walk. A third question is scheduling: because the registrar's grid fixes the class times, can the upper floors run on a shifted clock so that their peak does not coincide with the rest of the building?

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‡Students enrolled in COMP 363 during these two academic years are referenced here collectively.

The contribution is a transparent, reproducible model, not a measurement; the classical treatments of elevator traffic [2, 3] motivate its structure. All code, parameters, and result tables accompany this paper (Section 8).

2 Background

2.1 The building and its classrooms

We counted classrooms from the university's classroom A/V listing for Mundelein Center. Table 1 gives the count per floor. There are no classrooms on floors 8 and 9, so the one-bank limit at floor 9 splits the building cleanly into a lower group (floors 1–7, 45 rooms) and an upper group (floors 10–14, 9 rooms) that only the two-car bank can reach.

Table 1: Classrooms per floor in Mundelein Center.

Floor	1	2	3	4	5	6	7	10	11	12	13	14
Rooms	2	4	5	8	10	11	5	4	1	1	1	2

Floor 1 is an atrium lobby, and the stairs from floor 1 to floor 2 feel like two ordinary flights. Students leave the building through the lobby.

2.2 The scheduling grid

The Registrar's standard grid [1] defines seven 50-minute Monday–Wednesday–Friday blocks (8:15 to 3:35) and five 75-minute Tuesday–Thursday blocks (8:30 to 3:45) before the evening blocks, which start at 4:15 and are outside the scope of this study. Every consecutive pair of blocks, in either pattern, is separated by a 15-minute passing period. Our simulation models that period.

Classes seat 30–35 students, but enrollment is often 15–25. Elevator cars are rated for 3,000 lb, which is roughly 17–20 riders by weight; observed loads are four to six, so space, not weight, is the binding limit.

3 Model

3.1 Demand

One run simulates a single passing period. Time zero is the nominal end of the outgoing class and $T = 900$ seconds is the nominal start of the incoming class. Each classroom holds an outgoing and an incoming class independently with probability p_{occ}; each class has between 15 and 25 students, uniformly at random. Leavers depart uniformly over the first two minutes. A fraction q of leavers go on to another class in the building (assigned to an incoming room with probability proportional to remaining seats, and to an adjacent floor with probability a); the rest leave via the lobby. The seats in incoming rooms that transfers do not fill are occupied by students who arrive from the lobby with a triangular distribution over the ten minutes before class start, peaking three minutes before.

3.2 Elevators

Cars are simulated in one-second steps. The four-car bank serves floors 1–9 and the two-car bank floors 1–14. Passengers whose trip includes a floor above 9 can use only the two-car bank. Cars

follow a collective rule: continue in the current direction while there are riders to drop or calls ahead; otherwise reverse or go idle. Idle cars go to the nearest unclaimed call. Crucially, cars do *not* skip a floor when full: a car stops for any call in its direction of travel and then boards as many waiting people as it has room for. This reproduces the empty stops observed in practice. A car's capacity for a trip is drawn uniformly from $\{4, 5, 6\}$, and with backpack compliance b it is increased by two with probability b.

Timing assumptions are 3 seconds between floors (6 seconds from floor 1 to 2), 8 seconds of fixed door time per stop, and 1 second per rider boarding or alighting. These are assumptions, not measurements.

3.3 Behavioral measures and the offset

Walking. A rider whose trip starts on floor 2 or 3 and goes down walks with probability w_d (0.8 in the main experiments). A rider going up to floor 2 walks with probability 0.5 and to floor 3 with probability 0.2, reflecting the atrium. Walkers take 20 seconds per flight (40 for the atrium) and do not use the elevators. *Backpacks.* The compliance parameter b above, set to 1 in the main experiments. *Offset.* Classes on floors 10–14 start and end o minutes earlier than the grid ($o \in \{0, 5, 10\}$), preserving their length. A student moving from a lower floor to an upper floor therefore has $15 - o$ minutes between classes.

3.4 Metrics

For each run we record the mean and 90th-percentile wait (from when a rider is ready to when the rider boards), the share of students who reach their next class after it starts (walkers included), the share of stops at which nobody boards or alights, and the time to clear all queues. All figures average 60 replicates with different random seeds; we report standard errors of the mean for wait and lateness.

4 Results

The main experiments use $p_{occ} = 0.4$ (about 570 elevator riders per changeover), $q = 0.6$, and $a = 0.2$. We chose p_{occ} so that the baseline reproduces the qualitative description of a long line before every class without collapse; Section 4.4 shows how the results change with occupancy.

4.1 Behavioral measures

Table 2: Policies without an offset ($o = 0$). Wait is mean ± standard error; late is the share of students arriving after class start.

Policy	Mean wait (s)	P90 wait (min)	Late (%)	Empty stops (%)	Clear (min)
Baseline	314 ± 8	12.4	22.9 ± 1.4	14.5	25.0
Backpacks	217 ± 6	8.7	12.7 ± 1.2	9.3	20.2
Walking	294 ± 8	11.9	19.6 ± 1.3	13.9	24.6
Both	203 ± 6	8.5	11.0 ± 1.0	8.7	19.9

Table 2 and Figure 1 show the headline result. Holding backpacks in front reduces mean wait by 31% and nearly halves the share of late students. Walking is much weaker: it removes only about

7% of elevator riders, because floors 2 and 3 hold 9 of the 54 rooms and only partial compliance is realistic on the way up. Together the two measures reduce mean wait by 35%, to 3.4 minutes, and lateness from 22.9% to 11.0%. The effect of backpacks is nearly linear in compliance: each additional 25% of compliance removes roughly 20–30 seconds of mean wait (Figure 2).

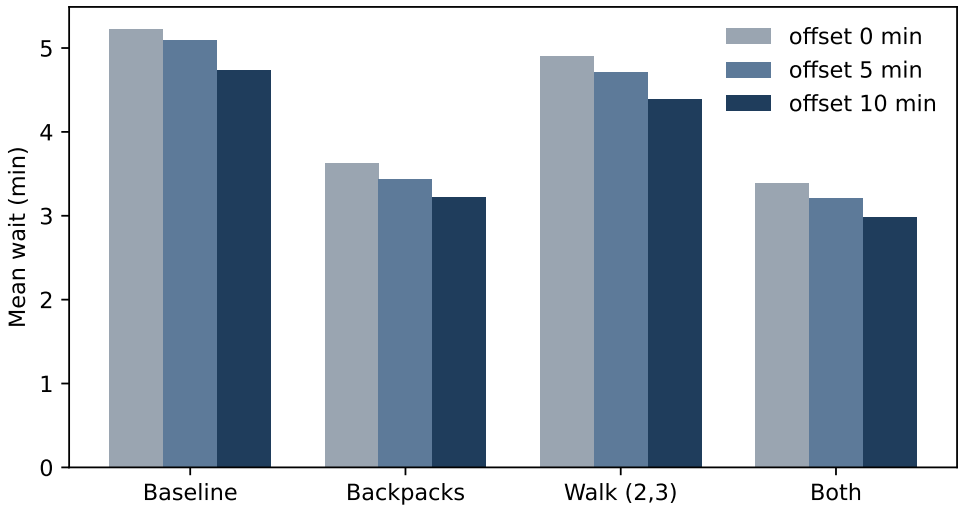


Figure 1: Mean wait by policy and offset of floors 10–14.

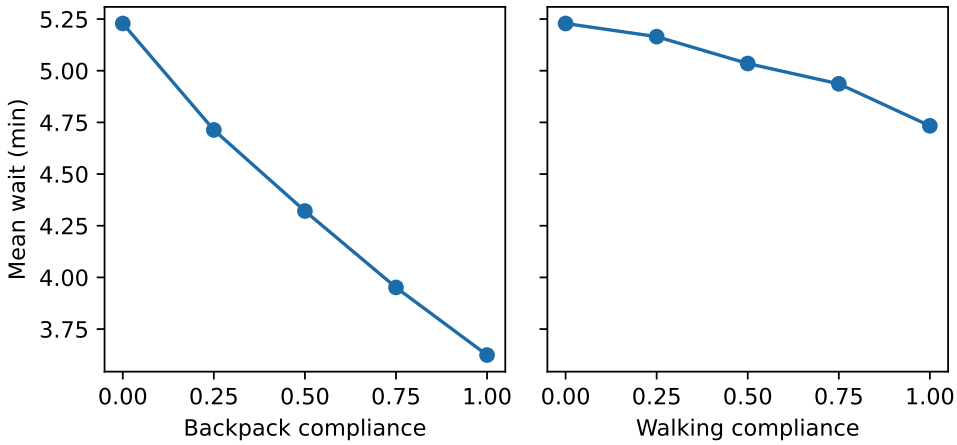


Figure 2: Mean wait as compliance grows. Left: backpacks. Right: walking (down-compliance shown; up-compliance scaled at 0.6 and 0.25 for floors 2 and 3).

Figure 3 shows why. Immediately after class ends, about 300 people are waiting in the baseline and the queue drains slowly; the arriving students then add to a backlog that has not cleared when the next class starts. The combined measures lower the peak and, more importantly, speed up the drain.

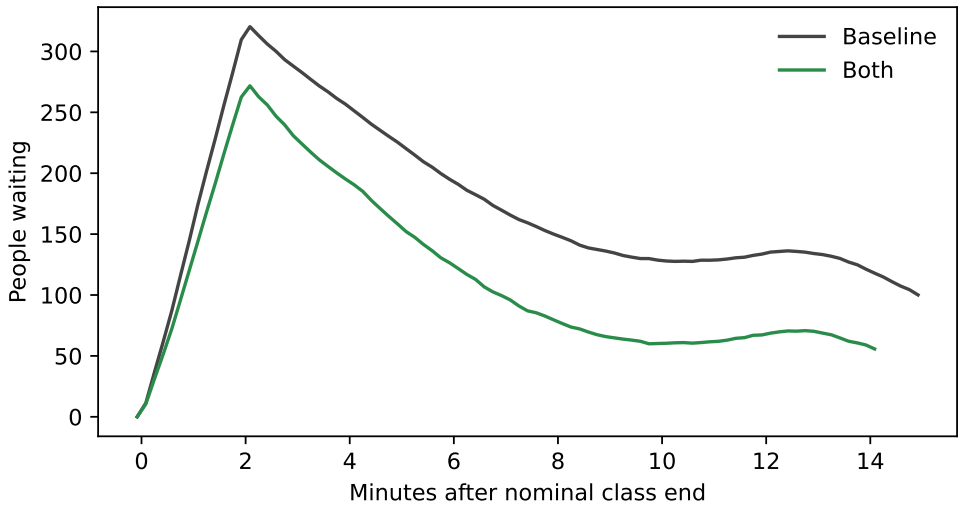


Figure 3: People waiting for an elevator over the passing period (mean of replicates, $o = 0$).

4.2 Where the waiting happens

Waits are not uniform across the building (Figure 4). Floor 10, the first floor served only by the two-car bank, is the worst: riders there wait about 10 minutes in the baseline and 7.5 minutes with both measures. Floors 5–7 are next, at about 5–6 minutes. The measures help everywhere but do not remove the upper-floor bottleneck, which comes from the structure of the banks.

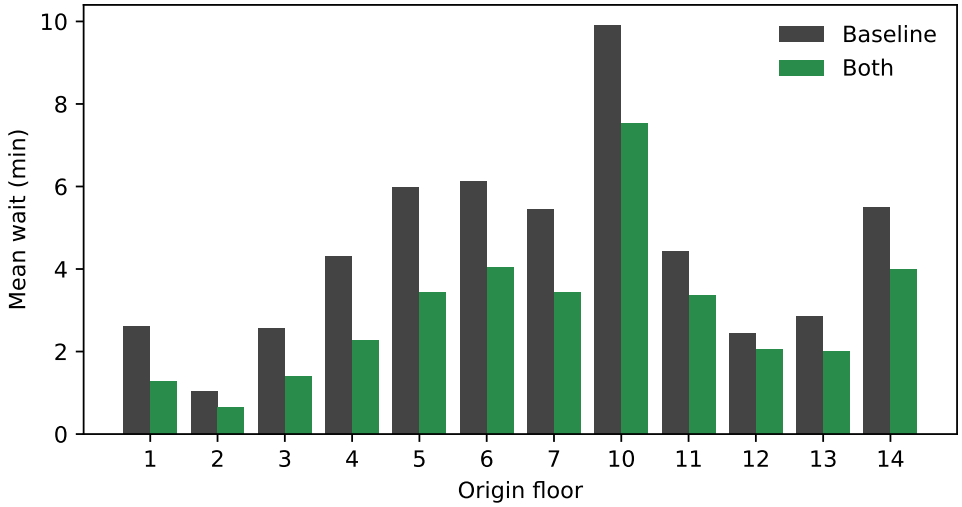


Figure 4: Mean wait by origin floor, baseline versus both measures ($o = 0$).

4.3 Time-multiplexing the upper floors

An offset trades one problem for another (Table 3, Figures 1 and 5). A 10-minute offset shortens the mean wait by about 9% in the baseline and 12% with both measures, and shortens the time to clear the queue by 22% and 20%. It does not reduce lateness: in the baseline the change is

Table 3: Effect of running floors 10–14 o minutes early.

Policy	Offset (min)	Mean wait (s)	Late (%)	Clear (min)
Baseline	0	314	22.9	25.0
Baseline	5	305	24.2	22.3
Baseline	10	284	23.2	19.5
Both	0	203	11.0	19.9
Both	5	192	12.5	17.5
Both	10	179	14.4	15.9

within noise, and with both measures lateness *increases* from 11.0% to 14.4%, because students moving from a lower floor to an upper one now have only five minutes. Students moving between adjacent floors are unaffected, since the group boundary falls between floors 7 and 10, which have no classrooms in between; the cost falls entirely on students changing between the lower and upper groups. A 5-minute offset gives about half of the reduction in clearing time at about half of the added lateness, so we regard it as the more defensible choice if the registrar would consider off-grid start times.

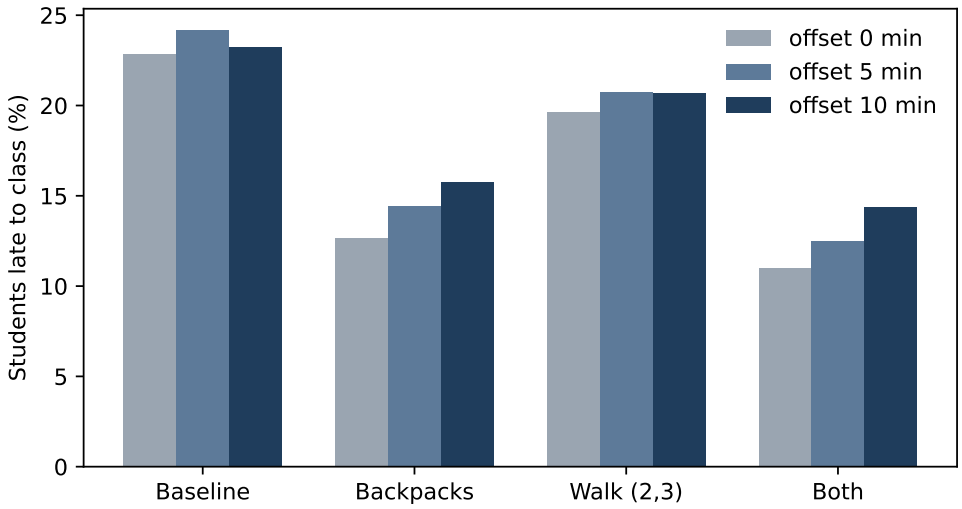


Figure 5: Share of students arriving after class start, by policy and offset.

4.4 Sensitivity

Occupancy. The absolute waits depend strongly on how many rooms are in use (Figure 6): from 2.6 minutes at 20% occupancy to 17 minutes at full occupancy in the baseline. The relative benefit of backpacks plus walking is stable, between 29% and 36% across the whole range. At 20% occupancy the baseline already has 7.9% late students, rising to 62% at full occupancy.

Door time. Doubling the door time per stop from 5 to 12 seconds raises the baseline mean wait from 236 to 432 seconds, while the relative benefit of both measures stays between 34% and 36%. Door time matters because every empty stop costs a full door cycle; this is the direct cost of the

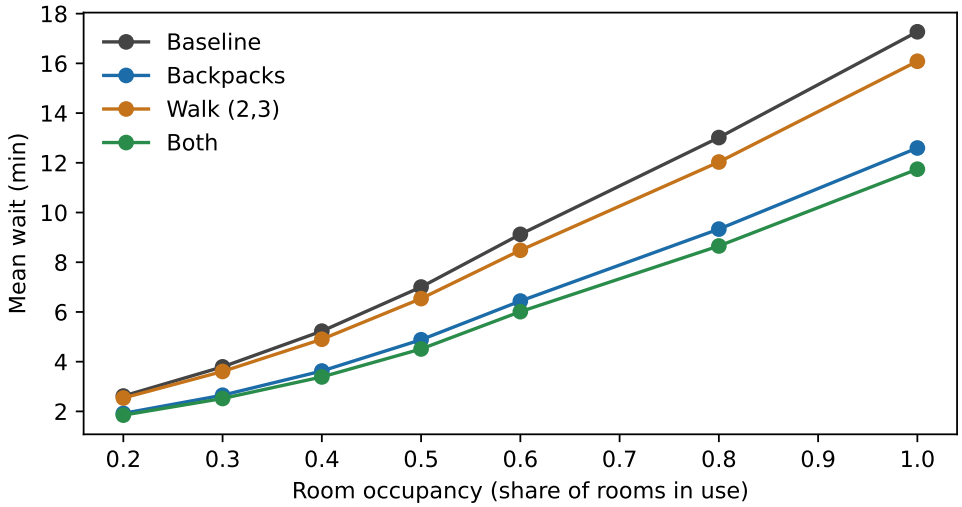


Figure 6: Mean wait as room occupancy grows.

lack of a full-car bypass.

Adjacent-floor transfers. Varying the share of transfers that go to an adjacent floor from 10% to 30% changes the late share by only 1–2.5 points in either direction (Figure 7), because without stairs the adjacent-floor riders still use the elevators. Offering stairs for one-floor trips, an extension beyond the two measures requested, reduces the combined-policy mean wait from 196 to 176 seconds (about 10%) at an adjacent share of 30% and 80% stair use, and the late share from 10.4% to 9.1%.

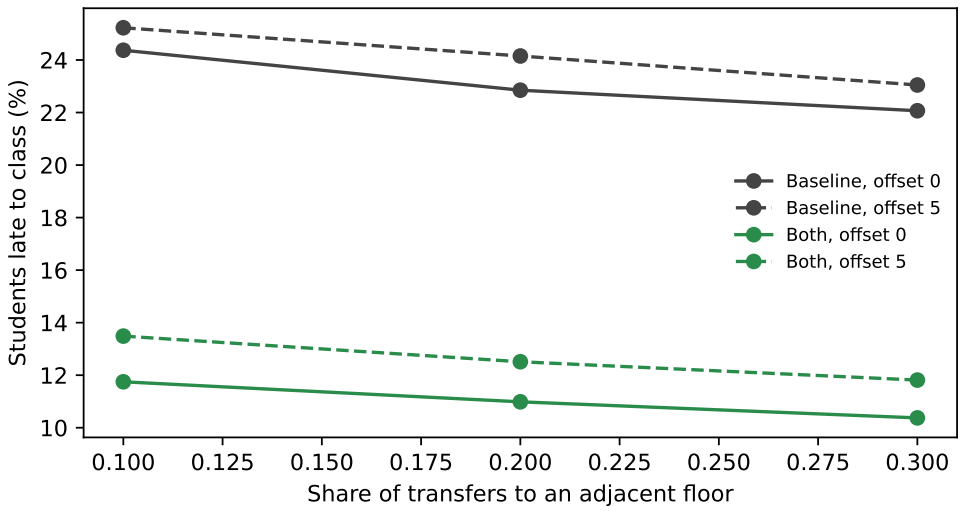


Figure 7: Share of late students against the adjacent-floor share of transfers.

5 Discussion

Backpacks matter most. The cars leave full because of volume, not weight. Making room for two more riders per car is by far the largest single lever in the model, and it costs nothing but a message. The model also shows that it scales smoothly: partial compliance gives partial benefit.

Walking is a supporting measure. Floors 2 and 3 are only 17% of the classrooms, and the atrium makes the climb to floor 2 feel like two flights. The gain comes mostly from the way down: walkers remove the empty stops on floors 3 and 2 that full cars make on their way to the lobby.

The upper floors need structural help. Floor 10 is the worst floor even after both measures, because all of its traffic goes through the two-car bank. Behavioral measures soften the problem; moving some of the load off the upper floors in the room assignment, or offsetting their clock, would act on the cause. The offset is a trade-off, not a free gain.

6 Limitations

- **No measurements.** Every timing parameter (travel, door, boarding) and every behavioral share (leaving spread, arrival distribution, transfer share, compliance) is an assumption. The model produces *relative* comparisons we believe are reasonable; absolute minutes and percentages should not be quoted as facts about Mundelein Center.
- **One passing period.** We simulate one changeover from an empty building. Backlog from the previous changeover would make the baseline worse.
- **Uniform schedule.** Without a room-by-block schedule we assume every room is in use with probability p_{occ} in every block and cannot distinguish heavy blocks from light ones.
- **Simple dispatching.** The real controller is proprietary and may behave differently. We deliberately did not model a full-car bypass or down-peak mode, because these were out of scope.
- **Static students.** Students do not adapt by leaving earlier when they see a line, which would reduce lateness in practice.

7 Conclusions and recommendations

1. Ask students to hold backpacks in front. In the model this is the largest single improvement: about 31% less mean wait and half the late arrivals at full compliance.
2. Ask students on floors 2 and 3 to walk, especially downward. It is a smaller gain, about 6% of wait on its own, but it combines with the first measure, and the two together reduce mean wait by about 35%.
3. If the registrar will consider it, try a 5-minute offset for floors 10–14. It shortens the time to clear the queue noticeably, but it costs students who move between floor groups, so measure before extending it.
4. Do not expect behavioral measures alone to fix floor 10. Its waiting time is structural.

5. Before quoting any number, run a timing study. Time round trips from floor 9, count stops and riders, and measure the door time per stop.

8 Reproducibility

The simulator (`model/elevator_sim.py`), the experiment runner (`model/run_experiments.py`), all result tables (`results/*.csv`), and these figures are in the project repository. Running `python run_experiments.py` with 60 replicates regenerates every table and figure.

References

[1] Loyola University Chicago, Office of the Registrar. *University Scheduling Grid*. `https://www.luc.edu/regrec/facultystaff/scheduling/universityschedulinggrid/`. Accessed 29 October 2026.

[2] G. C. Barney and S. M. dos Santos. *Elevator Traffic Analysis, Design and Control*. Peter Peregrinus, 2nd edition, 1985.

[3] G. R. Strakosch. *Vertical Transportation: Elevators and Escalators*. Wiley, 2nd edition, 1998.